

Proxemics-Based Interaction: Adapting CASTOR’s Behavior Through Proximity Sensing

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Abstract. Human–robot proxemics highlights the crucial role of interpersonal distance in shaping social engagement; guided by this perspective, this study evaluates the integration of proximity sensing technologies into a Social Assistive Robot (SAR) to improve interaction quality in human-robot interventions. Two sensor types, a laser-based Time-of-Flight sensor and a radiofrequency-based sensor were compared in terms of detection range, accuracy, precision, robustness, and ease of integration. The laser-based sensor demonstrated superior performance with a score of 91.03% and was embedded into the SAR. A perception protocol was developed to activate simple social behavior based on user distance, and the interaction was assessed with 12 neuro-typical adults, divided in two groups under autonomous and Wizard of Oz (WoZ) conditions. Results showed that the autonomous proximity-driven interaction encouraged users to approach the robot more closely and remain longer in socially meaningful zones, while maintaining comparable interaction quality to the WoZ condition. Perception scores were generally positive across both groups, with significant differences observed in Social Intelligence, Enjoyment, Perceived Trust, and Perceived Utility. These findings support the feasibility of incorporating presence-based functionality in SARs to facilitate more natural and engaging human–robot interaction, laying the groundwork for future use in interventions with children with Autism Spectrum Disorder (ASD).

Keywords: Social Robot · Human-Robot Interaction · Proximity Sensing.

1 Introduction

Social Assistive Robots (SARs) have gained significant attention in healthcare and therapeutic contexts due to their ability to engage users through natural and multi-modal interactions [1]. In interventions with children diagnosed with

ASD, these robots have shown potential to support the development of social, communicative, and emotional skills [2]. Their predictable behavior, physical embodiment, and capacity for social engagement make them especially suitable for children with Autism Spectrum Disorder (ASD), who often benefit from structured and consistent interactions [3], [4].

Detecting and responding to the presence of humans by proximity sensing is a crucial factor in making interactions with SARs meaningful, which serves as a primary mechanism to initiate social and context-aware behavior [1]. This aligns with research in human-robot proxemics, the area of Human-Robot Interaction (HRI) that examines how physical distance shapes social interaction among others. Prior work has shown that robots capable of adjusting their behavior based on interpersonal distance can create more socially appropriate interactions, as seen in autonomous proxemics-aware navigation systems [5] and studies analyzing how distance influences both physical and psychological aspects of HRI [6]. Similarly, Mitsunaga *et al* [7] demonstrated that adapting robot behaviors based on human nonverbal cues can enhance user comfort and engagement, such as interaction distance, gaze, and motion speed, especially when these adaptations are guided by subconscious feedback rather than explicit user input.

Beyond initiating interaction, proximity sensing serves as a complement to vision and touch in HRI, particularly in dynamic or overloaded environments, while vision systems are effective for long-range perception, they often struggle with occlusions and variable lighting [8]. Proximity sensors help address some of these limitations by enabling continuous, contactless detection, allowing robots to anticipate human presence even when visual or tactile cues are unavailable. This capability enhances both safety and responsiveness, which is valuable in therapeutic contexts where user comfort and social appropriateness are essential [1], [7]. In addition, proximity detection supports the regulation of interpersonal distance, helping robots respect intimate (0-45 cm), personal (45-120 cm), and formal (120-365 cm) zones - a key to maintaining user comfort [9].

Two of the most relevant technologies for proximity sensing in HRI are laser-based (optical) and radiofrequency-based (radar) sensors, each with distinct advantages [8]. Laser-based sensors, including reflected light intensity (O-RLI), time-of-flight (O-ToF), and triangulation types are widely used for their high spatial resolution, compact size, and fast response times. These features make them ideal for tasks such as manipulation, object detection, and collision avoidance in short-range, structured environments [10], [11]. On the other hand, radiofrequency-based sensors, especially Frequency-Modulated Continuous Wave (FMCW) radars, provide reliable performance regardless of lighting conditions or surface properties. They can simultaneously measure distance and velocity, penetrate certain materials, and operate without a direct line of sight [12], [13].

This study aims to evaluate and implement proximity sensing technologies for SARs, with a focus on improving HRI quality in intervention contexts. A direct and systematic comparison between laser-based and radiofrequency-based proximity sensors is conducted, assessing their sensitivity, range, precision, robustness, and suitability for integration into a SAR, thereby constituting the first

contribution: (1) a sensor-level evaluation for socially grounded interaction scenarios, a gap that remains insufficiently explored in therapeutic robotics. Based on these results, the most suitable sensor is integrated into a SAR, representing the second contribution: (2) a system-level integration within the CASTOR platform, ensuring compatibility with its ROS-based architecture and real-time interaction requirements. Finally, the integrated system is experimentally validated with healthy participants, forming the third contribution: (3) a user-level evaluation that demonstrates its effectiveness in detecting human presence and supporting proxemically appropriate robot responses. By linking sensor-level performance with interaction-level behavior, this work establishes a practical foundation for deploying proximity-aware SARs in future therapeutic applications, including interventions with children with ASD.

2 Methodology

This section outlines the framework used to compare two proximity sensor technologies for integration into a SAR in therapy, covering performance analysis, sensor integration, participant description, experimental procedure, and evaluation methods.

2.1 Proximity sensors

To evaluate the most suitable proximity sensing, was conducted a comparative study of two sensors: the VL53L3CX Time-of-Flight (ToF) (STMicroelectronics, Switzerland) laser-based sensor and the HLK-LD2420 24 GHz radiofrequency-based sensor (Hi-Link Electronic Co., Ltd, China). The VL53L3CX is a compact, ToF sensor. It operates using a 940 nm invisible class 1 laser emitter and a Single Photon Avalanche Diode (SPAD) array to measure the time it takes for emitted light to reflect off a target and return to the sensor [14]. This method allows for absolute distance measurement, regardless of the target’s color or reflectivity.

The HLK-LD2420 is a 24 GHz radar sensor developed for accurate human presence detection. It operates using Frequency-Modulated Continuous Wave (FMCW) technology, which emits continuous radio waves with linearly modulated frequency and measures the frequency difference between the transmitted and received signals. This corresponds to the time delay caused by the signal reflecting off a target and returning to the sensor [15]. This technique allows the sensor to detect motion, micro-motion, and stationary human presence.

2.2 Robot implementation

Robotic Platform: The CompliAnt Soft Robotic (CASTOR) is a cost-effective, open-source platform developed to support physical HRI in therapeutic environments for children with ASD [16]. Its structural design prioritizes safety, incorporating actuation mechanisms such as elastic joints and biologically inspired tendons. The robot features 14 degrees of freedom, powered by seven Dynamixel

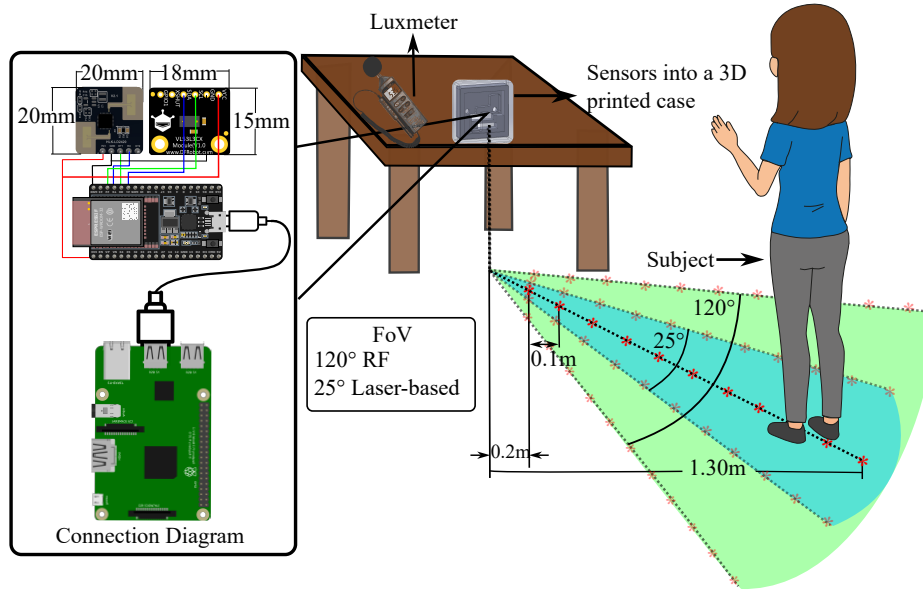


Fig. 1. Experimental setup illustrating the integration of the RF (24 GHz, 20 × 20 mm) and laser-based (15 × 18 mm) proximity sensors with a SBC, including the connection diagram, sensor housing, and the FoV and distance ranges used during the tests.

MX/AX motors (Robotis, France) and five MG996 servomotors (TowerPro, Taiwan), complemented by a passive damping system to absorb impact. In terms of interactive capabilities, CASTOR enables arm and neck movements, simulated emotional expressions, and a huggable system to encourage physical engagement. Its 3D-printed modular architecture ensures ease of assembly, maintenance, and customization.

Sensor Integration: Following the comparative evaluation of the proximity sensors, the selected one was integrated into the CASTOR robot to enable real-time detection of human distance. As illustrated in Figure 1, both sensors were initially connected to a microcontroller ESP 32 (Espressif Systems, China) and a single board computer (SBC) using serial communication. This approach was adopted due to the number of subsystems already integrated into the robot, which occupied the available I²C and UART communication ports, and to facilitate testing and data acquisition.

To integrate the sensor into the robot's structure, a 3D-printed piece was developed, designed to fit the dimensions of the selected sensor. This component aligns the Field of View (FoV) of the sensor with the intended interaction area of the robot. The sensor's data processing scripts were added to the CASTOR control architecture. The sensor's output was linked to the robot's behavior controller, allowing the system to create audio responses and antenna colors through

a Light Emitting Diode (LED) based on proximity, following the perception protocol.

2.3 Procedure

Comparative technologies: The two proximity sensors were tested under the same conditions. Distances were marked from 0.2 to 1.30 meters, and human approach trials were conducted in increments of 10 centimeters every 10 seconds as can be seen in Figure 1. For each sensor, detection readings were collected at angular positions matching their FoV: -12.5° to 12.5° for the laser-based (25° FoV) and -60° to 60° for the radiofrequency-based (120° FoV). In addition to range tests, precision was measured at each fixed distance by repeating presence and absence cycles. This helped calculate accuracy, precision, relative error rate, and precision rate. To see how each sensor worked in different lighting, the relative error rate was measured under normal, low, and high light levels. Light intensity was measured to keep conditions consistent. At each light level, tests were repeated from 0.2 to 1.30 m.

Perception protocol: The perception protocol was designed to activate simple social behavior from the robot based on proximity detection, using visual and auditory cues. It was structured into three proximity zones commonly detected in social interactions: intimate (≤ 45 cm), personal (45 - 120 cm), and formal (≥ 120 cm) [9]. Depending on the zone, the robot activated specific LED color cues and audio messages:

- Intimate: The robot turned on a green LED and initiated interaction with a greeting and follow-up questions. It continues with “Simon says” game, and finally says “Good bye”.
- Personal: The robot displayed a yellow LED and encouraged the person to come closer with phrases like “Come closer”. If the person remained at this distance after two attempts, it proceeded to say “Good bye”.
- Formal: A blue LED indicated the person stayed far. If the session was about to start, the robot invited the participant to approach. If it was the end, it concluded the interaction with “See you next time!”.

The evaluation of the perception protocol was conducted with 12 neurotypical adults aged between 20 and 30 years, all in healthy conditions. This work was approved by the ethics committee of the Federal University of Espirito Santo (85461424.7.0000.5542).

To evaluate how the proximity system influences user perception, participants were divided into two groups. One group interacted with the robot with the proximity feature enabled, while the other performed the same activity with the feature disabled. In the latter case, the robot’s behavior was controlled using a Wizard of Oz (WoZ) approach, disregarding the user’s distance. All sessions were video-recorded for subsequent behavioral analysis, and sensor data were collected to document the measured distances.

2.4 Evaluation variables

Comparative technologies: Each sensor was evaluated using a set of quantitative and qualitative metrics. The following variables were considered:

- Detection Range (DR): It was the distance at which the sensor reliably detected human distance across three angular zones: the central axis (0°) and the minimum and maximum FoV angles of each sensor in 10 cm increments. To quantify performance, the range was divided into the three proximity zones, each zone was evaluated based on the measurements with a relative error below 20%. If at least 80% of the values were below threshold, a point was assigned per valid zone, with 3 indicating detection across the entire range.

- Relative error rate (E%): This metric quantifies how far a sensor's output deviates from the actual distance value during presence detection. It is calculated as presented in Equation 1, where M represents the value from the sensor, and R corresponds to the reference.

$$Error (\%) = \left| \frac{M - R}{R} \right| * 100\% \quad (1)$$

- Precision rate (P%): To assess the consistency of sensor measurements at a fixed distance was used this standard metric as described in Equation 2, where σ denotes the standard deviation of the measurements, and μ represents their mean value.

$$Precision (\%) = \left| \frac{\sigma}{\mu} \right| * 100\% \quad (2)$$

- Precision and Accuracy - Classification-based (Pre CB and Acc CB): To evaluate detection performance in binary terms (presence vs absence), two metrics were calculated: precision and accuracy. These metrics are presented in Equations 3 and 4, respectively, where TP and TN denote correctly classified presence and absence cases, while FP and FN represent incorrect classifications.

$$Precision = \frac{TP}{TP + FP} \quad (3)$$

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (4)$$

- Robustness to environmental conditions (RE): This was assessed by measuring the relative error rate at three controlled lighting conditions: low (RE Ll), normal (RE Nl), and high luminosity (RE Hl). Ambient light levels were measured using a lux meter (DT-8820, CEM, China).

- Ease of Integration (EI): This qualitative metric reflects how easily each sensor could be embedded in the SAR, considering three factors: Mechanical compatibility (MC), Software compatibility (SC) and Hardware complexity (HC). The score was calculated using the weighted Equation 5.

$$EI = 0.4MC + 0.35SC + 0.25HC. \quad (5)$$

Each component was rated on a Likert scale from one to five by the investigators, where higher values indicate better compatibility or lower complexity, and lower values indicate incompatibility or higher complexity. The weighting coefficients were defined according to the impact of each factor on real-world integration into the CASTOR platform. MC was prioritized (0.4) due to physical integration constraints, followed by SC (0.35), given its influence on communication with the ROS-based architecture. HC (0.25) was assigned a lower weight, as it can be partially mitigated through design adaptations.

- Sensor Selection Score (SS): To determine the most suitable sensor for this application, the Equation 6 was used. The evaluation variables were weighted according to their relevance to this social HRI. The variables are: E%, P%, EI, Acc CB, Pre CB, Rmax, RE Ll, and RE Hl.

$$SS = 0.025((1 - E\%) + P\% + (1 - RELl)) + 0.2(EI + AccCB + PreCB) + 0.1625(Rmax + (1 - REHl)). \quad (6)$$

Higher weights (0.2) were assigned to EI, Acc CB, and Pre CB, as reliable perception and seamless integration are critical for stable and meaningful HRI. Range-related factors (Rmax and RE Hl) were given intermediate weights (0.1625), reflecting their importance in proxemics-based interaction. Lower weights (0.025) were assigned to secondary indicators (E%, P%, and RE Ll), ensuring they contribute to the evaluation without disproportionately influencing the final score. This weighting strategy provides a balanced multi-criteria assessment aligned with the system requirements and experimental objectives.

Perception protocol: To evaluate the acceptance of the robot’s proximity-based interaction protocol, the Unified Theory of Acceptance and Use of Technology (UTAUT) framework combined with perception metrics was applied [16]. This perception questionnaire uses 13 metrics (Friendliness (FRI), Safety (SFT), Social Intelligence (SocInt), Enjoyment (ENJ), Naturalness (NAT), Social Influence (SocInf), Social Presence (SP), Perceived Ease of Use (PEOU), Attitude Toward Technology (ATT), Robot Safety (RobSFT), Perceived Trust (PT), Perceived Utility (PU), and Anxiety (ANX)). The participants had the interaction randomly with the autonomous-based protocol or with the WoZ. After the interaction participants completed the questionnaire to capture their perception of the system’s usability and effectiveness, comparing between the groups. The survey also includes two open questions that describe the robot and the interaction with three adjectives [17].

To examine statistical differences between groups, each construct was first tested for normality using the Shapiro–Wilk test with a 5% significance threshold. Constructs that met the normality assumption were analyzed using an independent-samples t-test. Constructs that did not meet normality criteria were evaluated using the non-parametric Wilcoxon rank-sum test. In both tests, a one-tailed comparison was applied, using a “greater-than” alternative hypothesis to determine whether one interaction condition yielded higher scores than the

other. This analytical approach ensured that each construct was evaluated with the most appropriate statistical method according to its distribution.

In addition to subjective measures, a behavioral analysis was conducted using the recorded distance and video data. These materials were used to determine the duration of each interaction and to assign a quality rating (0–5) in 15-second intervals throughout the session. This procedure allowed for an objective comparison of user behavior and interaction dynamics across the two experimental conditions. To ensure the reliability of the qualitative coding, two independent researchers annotated all interaction intervals. Inter-rater reliability (IRR) was assessed using a two-way random-effects model with absolute agreement for single measurements called Intraclass correlation coefficient (ICC(2,1)), which is an approach for behavioral coding [18].

In addition to reporting the ICC estimate, the analysis includes the associated p-value and 95% confidence interval. The p-value is obtained through an F-test that evaluates whether the observed agreement between raters is significantly greater than zero, based on the ratio of mean squares derived from a two-way Analysis of Variance (ANOVA). Confidence intervals are computed using the noncentral F-distribution, providing a range of plausible ICC values and accounting for the variability in the reliability estimate. These statistical outputs support a more robust interpretation of coder agreement and strengthen the validity of the behavioral annotations.

3 Results

The comparative analysis of the proximity sensors, summarized in Table 1, highlights the performance of the laser-based sensor across most evaluation variables. It achieved a broader DR (84.72%) compared to the RF sensor (5.55%), and a notably lower E% (6.29% vs. 16.54%). The laser sensor also demonstrated higher P% (65.03%) and Acc CB (98.19%) while maintaining 100% in Pre CB, as well as the RF sensor.

Table 1. Comparative Analysis of Evaluation Variables.

Variables (%)	RF Sensor	Laser Sensor
DR	05.55	84.72
E	16.54	06.29
P	49.32	65.03
Pre	100.0	100.0
Acc	94.94	98.19
RE L1 (97.20 lux)	19.39	08.07
RE H1 (339.2 lux)	20.56	11.03
RE N1 (237.6 lux)	16.54	06.29
EI	72.00	84.50
SS	72.53	91.03

The results indicate clear performance differences between the two sensors. The RF sensor showed substantial variability across the tested distances and FoV angles, with many measurements presenting high dispersion and E% values exceeding the 20% threshold. In contrast, the laser-based sensor maintained a consistently tight distribution of errors, with E% remaining mostly below the threshold. Under varying lighting conditions, the laser sensor also demonstrated greater robustness, yielding RE values below 12%, whereas the RF sensor produced higher RE rates ranging from 16% to 21%. Additionally, the laser sensor achieved a higher EI score of 84.5%, resulting in an overall SS of 91.03%, compared to the RF sensor's 72.53%.

Figure 2 presents the integration of the proximity sensor into the SAR, emphasizing the custom-designed piece that securely the sensor and aligns its FoV with the user's frontal area. This enabled a reliable and consistent detection of the user's distance in real time. The figure also defines the three operational zones: intimate, personal, and formal, which were essential for adapting the robot's behavior.

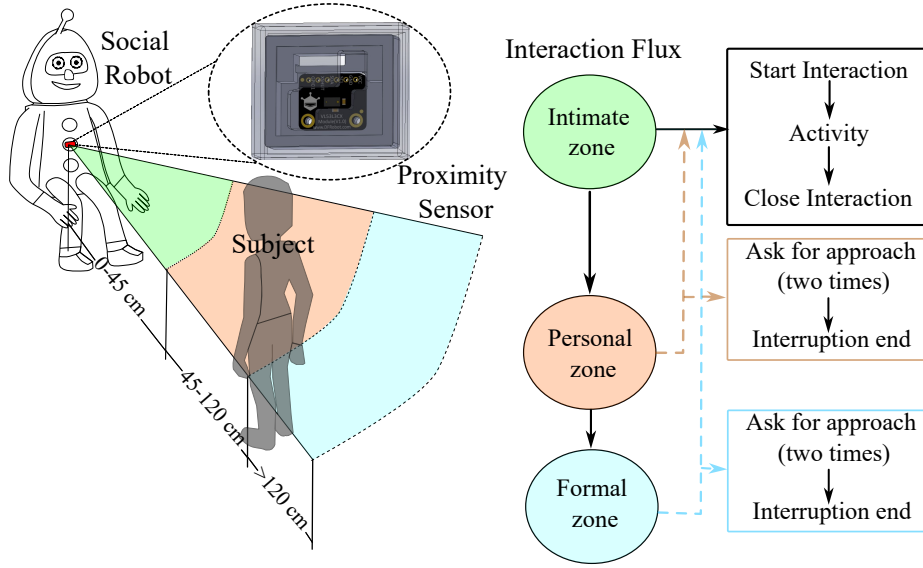


Fig. 2. Integration of the sensor into the SAR, showing the piece and the defined interaction zones. Also illustrates the interaction flux.

As shown in Figure 3, the perception scores for the autonomous interaction and WoZ condition were compared across 13 evaluated constructs. Overall, both groups reported generally positive impressions of the robot, with high scores across most metrics. However, four constructs showed statistically significant differences between conditions. SocInt was rated significantly higher in the WoZ condition compared to the autonomous interaction ($p = 0.03445$), indicating that

participants perceived the robot as more socially competent when its behavior was teleoperated. ENJ also favored the WoZ condition ($p = 0.04401$), suggesting that participants experienced a more pleasant interaction in the human-operated scenario. Additionally, PT ($p = 0.01044$) and PU ($p = 0.03834$) were significantly higher in the WoZ condition, reflecting greater confidence in the system and a stronger sense of usefulness when the robot was controlled by a human operator. All other constructs, including FRI, SFT, NAT, SocInf, SP, PEOU, ATT, RobSFT, and ANX, showed no significant differences between conditions, indicating broadly similar and favorable perceptions of the robot regardless of the interaction modality.

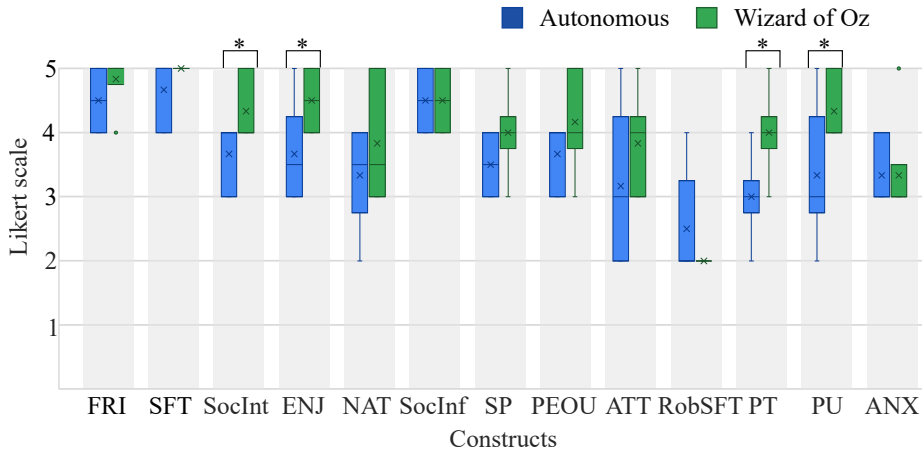


Fig. 3. Survey outcomes of perception of FRI, SFT, SocInt, ENJ, NAT, SocInf, SP, PEOU, ATT, RobSFT, PT, PU, ANX comparing the proximity-based autonomous and the WoZ interaction. * indicate statistically differences between conditions ($p\text{-value} < 0.05$).

Table 2 summarizes the proximity-related behavioral metrics for the WoZ and autonomous interaction groups. Clear differences emerged between conditions. Participants in the WoZ group maintained a larger average distance from the robot (89.12 ± 11.48 cm) compared to those in the autonomous condition (42.51 ± 4.58 cm). The WoZ group also achieved a greater minimum distance (71.97 ± 18.61 cm) from the robot compared to the autonomous group (22.70 ± 8.28 cm). Analysis of interaction zones highlighted contrasting behaviors: participants in the autonomous condition spent a considerable amount of time within the robot’s intimate zone (97.67 ± 35.58 s), whereas this zone was never entered during WoZ interactions. In contrast, the WoZ group spent significantly more time in the personal zone (166.33 ± 62.51 s) and a measurable amount of time in the formal zone (10.83 ± 24.22 s), while participants in the autonomous condition did not engage in the formal zone at all. These results indicate that the proximity-driven

autonomous behavior encouraged participants to engage more closely with the robot, reducing interpersonal distance compared to the manually controlled WoZ condition.

Table 2. Proximity metrics comparing the WoZ control group and the autonomous proximity-driven group.

Metric	WoZ (Mean $\pm$ SD)	Autonomous (Mean $\pm$ SD)
Distance (cm)	89.12 $\pm$ 11.48	42.51 $\pm$ 4.58
Minimum distance (cm)	71.97 $\pm$ 18.61	22.70 $\pm$ 8.28
Time in intimate zone (s)	0.00 $\pm$ 0.00	97.67 $\pm$ 35.58
Time in personal zone (s)	166.33 $\pm$ 62.51	64.67 $\pm$ 50.89
Time in formal zone (s)	10.83 $\pm$ 24.22	0.00 $\pm$ 0.00

Distance-over-time patterns for participants in the WoZ and autonomous conditions are shown in Figure 4. The shaded regions correspond to social distance zones (intimate, personal, and formal), providing context for interpreting participants’ proximity to the robot throughout each session. In the WoZ example, the participant maintained a relatively stable distance within the personal zone, with limited movement toward closer zones. In contrast, autonomous example shows more variation in proximity, including periods in the intimate zone and the minimum distance reached during the interaction.

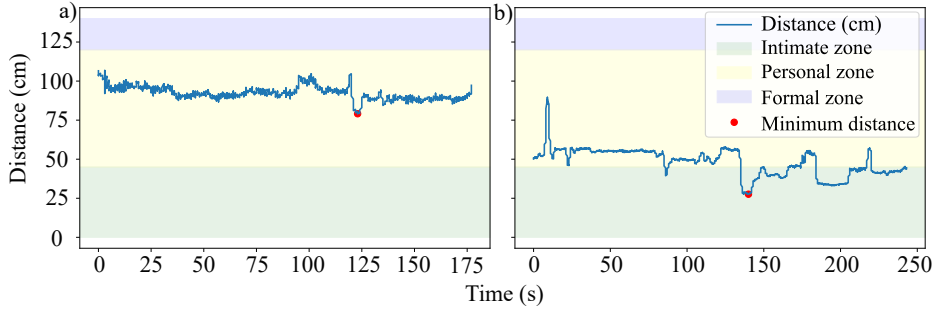


Fig. 4. Examples of distance-over-time curves for participants: a) WoZ group and b) autonomous conditions. Shaded areas represent social distance zones.

Finally, for the behavioral analysis, the IRR between the two independent researchers demonstrated a good level of agreement, with ICC = 0.738, 95% Confidence Interval (CI) = [0.44, 0.89], and p-value = 0.0001, indicating statistically significant consistency across all annotated sessions. Based on these reliable annotations, interaction quality was examined across distance zones. As shown in Table 3, the global mean quality scores were similar for the intimate

(3.898 ± 1.303) and personal (3.838 ± 1.375) zones, while formal-distance intervals—although infrequent—received the highest average rating (4.7 ± 0.632). When examining the distance zones, the WoZ condition showed its highest interaction quality in the formal zone, followed by the personal zone. In contrast, the autonomous condition exhibited its highest quality ratings in the intimate zone, with lower average scores in the personal zone and no intervals recorded in the formal zone. These patterns indicate that, in both conditions, interactions tended to be rated more positively when participants were physically closer to the robot.

Table 3. Interaction quality by distance zone, with global descriptive statistics and group-level mean ratings.

Distance Zone	Global (Mean $\pm$ SD)	Count	Autonomous Group (Mean)	WoZ Group (Mean)
Intimate	3.898 ± 1.303	4439	3.898	-
Personal	3.838 ± 1.375	8268	3.250	4.193
Formal	4.7 ± 0.632	10	-	4.7

4 Discussion

The RF sensor offers a low cost (approximately 2 USD), compact design, and the advantage of non-line-of-sight operation, making it an option for deployment in IoT and robotics applications [19]. However, while RF sensors can detect the presence of an object within a given range (Acc and Pre CB $\geq 94\%$), they do not provide precise distance measurements [20]. This limitation is consistent with the findings of this study, where the RF sensor achieved a P% of 49.32% and an E% of 16.54%, indicating limited reliability for accurate proximity-based interaction. In contrast, the laser-based sensor offers greater functional flexibility, is lightweight, highly accurate, and demonstrates strong reliability [21]. These advantages are consistent with the results obtained in this study, where the laser sensor achieved a DR of 84.72%, a low E% of 6.29%, and a high P% of 98.19%. Additionally, laser-based systems can provide wider bandwidth, lower power consumption, and an enhanced potential for covert data transmission compared to radio-based alternatives [22].

The proximity-related behavioral results reveal differences between the WoZ and the autonomous conditions as can be seen in Table 2. Participants consistently interacting with autonomous proximity-driven behavior and remained closer to the robot, as reflected in the shorter average and minimum distances, and in the time spent in the intimate zone. This contrasts with the WoZ condition, where participants maintained larger interpersonal distances and never entered the robot’s intimate space. These patterns align with prior work in robot proxemics showing that autonomous behaviors that adjust movement respon-

sively tend to invite closer engagement and modulate users’ comfort with distance [5]. The distance-over-time curves in Figure 4 reinforce these behavioral distinctions, with autonomous interactions displaying richer variability and more transitions across zones, while WoZ interactions remained more static and confined to the personal zone.

The inter-rater reliability ($ICC = 0.738$) establishes that the behavioral coding was consistent across researchers, supporting the validity of subsequent interaction-quality comparisons. Based on these annotations, Table 3 shows that interaction quality followed a similar trend in both conditions: closer distances were associated with higher ratings, with the intimate zone yielding the most positive evaluations. This finding is consistent with proxemics-based HRI research demonstrating that reduced interpersonal distance can strengthen perceptions of engagement and social presence [6].

The results of the user interactions showed generally positive evaluations across perception metrics in both the autonomous and WoZ conditions. However, four constructs: SocInt, ENJ, PT, and PU, had significantly higher scores in the WoZ condition. This suggests differences in social and experiential perceptions between human-operated and autonomous robot behavior. These differences may be partly explained by the nature of the interaction tasks. The interaction was dynamic, and the robot issued instructions such as turning around or performing a squat. During these movements, participants frequently crossed between social distance zones, which caused the proximity sensor to register rapid changes and trigger additional prompts for the participant to move closer to the robot. These repeated adjustments may have contributed to the perception that the robot’s autonomous behavior was less socially competent or less able to interpret the situation appropriately. Additionally, the small sample size and the limited scope of the tasks—focused mainly on following simple physical commands—may also have influenced the perception differences observed. For the remaining constructs, no significant differences were found, suggesting that the autonomous proximity-based protocol was perceived as comparable to the WoZ condition for most dimensions of interaction quality. These results support the viability of integrating autonomous proximity-based functionalities into the CASTOR robot while also improving areas such as SocInt, ENJ, PT and PU where further refinement may enhance the autonomous system’s perception. As CASTOR is primarily intended for interaction with children, future evaluations with the target population may provide additional insights and potentially higher perception ratings in these domains.

Building on these findings, future work will focus on expanding the sample size and including participants from the intended target population to strengthen the generalizability of the results. Additionally, upcoming iterations will explore more flexible interaction protocols and evaluate the system in real-world therapeutic settings, enabling a more robust assessment of its applicability and adaptability.

5 Conclusions and Future Work

The comparative evaluation of the laser-based and RF proximity sensors demonstrated the laser sensor's superior performance for integration into SARs. According to the defined evaluation criteria, the laser sensor obtained a higher SS of 91.03%, compared to 72.53% for the RF sensor, justifying its selection for integration into the robotic platform.

Moreover, the integration of proximity-based functionalities into the CAS-TOR robot demonstrated promising results, enabling socially appropriate interactions that were perceived as effective and natural by the participants, who spent more time in socially meaningful zones, such as intimate and personal, without reducing interaction quality. Together with the strong IRR with ICC = 0.738 and the fact that both autonomous and WoZ interactions were well received, the findings support the feasibility of autonomous presence detection for social engagement.

An additional advantage of the proximity-based approach is that it removes the need for an external person to teleoperate the robot during the session, which can simplify the interaction setup and reduce cognitive load for the operator when manual control would otherwise be required. However, the study need to improve aspects such as SocInt, ENJ, PT, and PU. Future work should focus on refining adaptive behaviors, validating the system with the target population, and implementing the protocol in real therapeutic settings.

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